Alem — participant handout

You are an agent in a 3-agent cooperative survival game. Your goal is to gather resources, craft gear, fight monsters, and descend through 9 dungeon levels, while coordinating with teammates. You must survive — if your health reaches zero, you die, and if all agents die the game ends. Maximize the number of achievements while staying alive.

How to play

- Each turn, choose exactly one action.
- **Movement** uses absolute directions: north, south, east, and west. Any move attempt changes your facing to that direction, even if the move is blocked and you stay in place. A move is blocked if the target tile is solid, including trees, stone, ore veins, walls, crafting stations, chests, and plants, or if it contains water, lava, a mob, or another player. If repeated move attempts do not change your position, that direction is blocked. You can also use a blocked move to turn in place, for example to face an adjacent tree.
- **Facing**: your facing direction is set by your last movement action and persists until you move again. **Do** always targets the tile in your current facing direction.
- **Do** is your main interaction: face a tile and use the **Do** action on exactly that tile to chop trees, mine ore, attack creatures, drink water, open chests, or revive a downed teammate. If the faced tile contains a downed teammate, Do revives them. If the faced tile contains a living teammate, Do targets that teammate instead and can cause friendly fire. For synchronous-style coordination, all required agents must stand next to the same target tile, face it, and act together.
- **Crafting**: stand next to (including diagonally) the required station and use the craft action; you do NOT need to face it. Diamond items always require an adjacent epic forge, not a table.
- **Placing**: face the target tile, then use the place action. Tables and furnaces need an empty non-solid tile that is not water or lava; stone can also be placed into water (costs 1 stone). Place Plant puts a sapling on the faced tile. Place Torch lights dark areas.
- **Ranged combat**: use Shoot Arrow while facing a creature (requires a bow + arrows). Bows are found in dungeon chests.
- **Elite mobs** are tougher and deal more damage; coordinating with teammates (multiple agents attacking together) makes them much easier to defeat.
- **Request/Give**: use Request [Resource] to broadcast a resource request to teammates for 10 turns; teammates can use Give to Agent X to transfer one unit of the requested resource directly — no adjacency required, works at any distance. Give only appears as an available action when a teammate has an active Request.

Survival stats

Food, drink, and energy deplete gradually over time — roughly every 20-30 steps you lose 1 point of each (dexterity slows this rate). When food or drink reaches 0, your health starts dropping. When energy reaches 0, you automatically fall asleep and cannot act until energy is full. While sleeping, you take 2.5x damage from all sources. Mana does NOT decay — it is only spent by casting spells or enchanting. Mana slowly regenerates over time (faster while sleeping). - **Sleep**: choose this voluntarily to recover energy at 2x the passive rate. Ends automatically when energy is full. - **Rest**: choose this to recover health gradually. Requires food, drink, and energy all > 0; ends when health is full or a stat runs out.

Roles

Role-restricted actions succeed with reduced probability for non-specialists. Depending on the difficulty configuration, non-specialist success rates are 10%, 40%, or 70%. Specialist success rate is always 100%.

- **Forager**: collecting water, saplings, eating passive mobs (e.g. cows/bats/snails). Also has 3x base food and drink capacity.
- **Miner**: crafting pickaxes/torches, placing stone.
- **Warrior**: crafting swords and arrows. Also deals 2x melee damage, and specializes in enchanting swords and bows.
- No role restriction: Place Table, Place Furnace, Wood Sword, Iron Armour, Diamond Armour

Coordination

Some tasks, tiles, creatures, and structures require multiple agents. When collaboration is possible the observation will include a short coordination hint. Follow the rules below to coordinate safely and efficiently.

- **Sync**: N agents must each stand on a different tile adjacent to the shared target, each facing it, and all choose the **Do** action on the same turn. Approach from different sides so every agent targets the shared tile directly. If a teammate is between you and the target, your **Do** will hit the teammate instead and can cause friendly fire.
- **Handover**: one agent starts the task with **Do** (pays the resources required), and another agent finishes it with **Do** within a small time window shown in the observation. If no agent completes it in time, the site resets and materials are refunded to the initiator. Follow the exact handover timing when specified.
- **Construction**: Construction sites (shelters, forges, beacons) may require either sync or handover. Always follow the coordination rule shown in the observation for that site.
- **Elite mobs**: stronger enemies may benefit from or require coordinated attacks. Attack from different sides, avoid standing between a teammate and the mob, and avoid blocking another agent's attack.
- **Revive**: Face a downed teammate and use **Do** to revive them.
- **Epic forge / Diamond crafting**: Diamond-tier items require multiple agents to craft simultaneously at an adjacent epic forge. All required agents must choose the crafting action for the same item on the same turn while adjacent to the forge.

Resource chain

Trees → wood (no tool required) → Stone/Coal (needs wood pickaxe) → Iron (needs stone pickaxe) → Diamond (iron pickaxe) → Ruby/Sapphire (diamond pickaxe)

Crafting recipes

All recipes consume the listed materials. Stations: Table (2 wood), Furnace (1 stone) - Wood pickaxe/sword: table + 1 wood - Stone pickaxe/sword: table + 1 wood + 1 stone - Iron pickaxe/sword: table + furnace + 1 wood + 1 stone + 1 iron + 1 coal - Iron armour: table + furnace + 3 iron + 3 coal - Diamond pickaxe: epic forge + 1 wood + 3 diamond + enough agents crafting the same item there on the same turn - Diamond sword: epic forge + 1 wood + 2 diamond + enough agents crafting the same item there on the same turn - Diamond armour: epic forge + 3 diamond + enough agents crafting the same item there on the same turn - Arrows: table + 1 wood + 1 stone (yields 2) - Torch: table + 1 wood + 1 coal (yields 4)

Construction (at a construction site, face it and use Build action): - Build Shelter: needs 10 wood + 5 stone. Shelters result in +50% energy regeneration while resting (all agents). - Build Forge: needs 10 stone + 3 iron + 2 coal.

Creates an epic forge, which enables diamond gear crafting. - Build Beacon: needs 3 iron + 2 coal. Expands the lit area on this level.

Attributes

Gain 1 XP each time you descend to a new floor. Spend XP with Level Up actions. - **Strength**: max health = 8 + strength - **Dexterity**: max food = 7 + 2*dexterity (+2 extra for foragers); max drink = same; max energy = 7 + 2*dexterity - **Intelligence**: max mana = 6 + 3*intelligence; enchantment damage +5% per point above 1.

Progression

1. Gather wood → place a table → craft a wood pickaxe; craft a wood sword early if combat is likely.
2. Mine stone and coal → place a furnace → craft iron tools and iron armour.
3. To descend: stand on the `ladder_down` tile (visible in your observation when close) and use the Descend action. The ladder only becomes usable after enough monsters on that level have been killed. Only one agent needs to use Descend/Ascend — all teammates are teleported with them.
4. In dungeons, open chests for loot such as bows, potions, and spell books, and mine gems as your tools improve.
5. Enchant weapons, bows, or armour at enchantment tables by spending 9 mana and 1 gem; ruby gives fire, sapphire gives ice.
6. Craft diamond gear at an epic forge, coordinating with teammates when required.
7. Repeat across all dungeon levels until the final boss.

Controls (keyboard)

Players take turns on one keyboard: the player named in the box at the top-left of the map (and boxed in the top bar) acts, then the next. The world advances once every player has chosen. The map is centred on you and framed in your colour. - Move / turn: W A S D (north / west / south / east). Moving into a blocked tile just turns you to face it. The line under the map always tells you what you are facing and what it needs. - Do / interact: Space — acts on the tile you face (chop, mine, attack, drink, open chest, revive a downed teammate) - Sleep: Tab (regain energy) Rest: E (regain health) Skip turn: Q - Descend / Ascend a ladder: . / , (the whole team moves together; the ladder down opens after 8 monster kills on the floor) - Place: R Stone · T Table · F Furnace · P Plant sapling · J Torch - Craft pickaxe (next to a table): 1 Wood · 2 Stone · 3 Iron (+furnace) · 4 Diamond (epic forge, together) - Craft sword (next to a table): 5 Wood · 6 Stone · 7 Iron (+furnace) · 8 Diamond (epic forge, together) - Craft armour: Y Iron · U Diamond Craft: O Arrows · [Torches - Fight: I Shoot arrow (needs a bow from a chest + arrows) · G Cast spell (after reading a book) - Potions: Z Red · X Green · C Blue · V Pink · B Cyan · N Yellow (each colour's effect is random per world; test when safe) - Magic: M Read book · K Enchant sword · L Enchant armour · ; Enchant bow (at an enchantment table, costs 9 mana + 1 gem) - Level up (spends 1 XP, earned per new floor):] Dexterity · - Strength · = Intelligence - Ask the team for a resource: F1 Food · F2 Drink · F3 Wood · F4 Stone · F5 Iron · F6 Coal · F7 Diamond · F8 Ruby · F9 Sapphire. Teammates see it for 10 steps (press again to renew). - Give 1 unit to a teammate who has an open request: Backspace gives to the lower-numbered other player, Delete to the higher-numbered one (P1: Bksp→P2, Del→P3 · P2: Bksp→P1, Del→P3 · P3: Bksp→P1, Del→P2). The sidebar names who each key reaches and draws it white when it would work. - Build at a construction site (face it): 9 Shelter · 0 Forge · ` Beacon

Reading the screen

- Sidebar: every key, drawn white when that action would work right now and faded when it would not; the TEAM block shows each player's health and open request and which Give key reaches them.

- Map borders: green = soft coordination (works alone, bonus together) · cyan = hard coordination (fails alone) · orange dashed = handover (one starts, another finishes in time) · the number is how many players it needs · red outline + gold tint = elite creature (the facing line says whether it is soft or hard).
- Bottom-left: recent rewards, damage and achievements.

During the session

- ENTER starts each episode · ESC opens the pause menu (resume / save / reload / quit) · close the window to end the session.
- New achievements pop up on screen as you unlock them — unlocking as many as possible (while staying alive) is the goal.